Sales Performance Analytics Report

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Executive Summary

This project analyzed e-commerce sales data to identify factors influencing revenue generation and customer purchasing behavior. Microsoft Excel was used for data cleaning, KPI calculations, and PivotTable analysis, while Tableau was used to develop an interactive dashboard.

The analysis found that Electronics generated the highest overall revenue, customers aged 55+ contributed the largest share of sales, and revenue exhibited seasonal fluctuations throughout the year. These findings support data-driven decisions related to marketing strategy, inventory planning, and customer engagement.

Project Objective

The purpose of this project was to analyze sales data and examine revenue across products, customers, cities, time periods, and payment methods.

Business Questions

The analysis wanted to answer the following questions:

- Which product categories generate the most revenue?
- Which products are the top performers in terms of revenue?
- Which cities generate the highest revenue?
- How does revenue vary across customer age groups?
- How does revenue change throughout the year?
- What payment methods are most commonly used by customers?

Dataset Overview

The dataset used in this project was obtained from Kaggle and represents transactional sales data for a fictional e-commerce company. The dataset includes customer demographics, information on products, order details, payment methods, and customer review scores.

Variables Used

Variable	Description
customer_id	Unique identifier assigned to each customer who placed an order
order_date	Date on which the order was placed
product_id	Unique identifier assigned to each product

category_id	Unique identifier assigned to each product category.
category_name	Name of the product category.
product_name	Name of the product purchased.
quantity	Number of units purchased in the order.
price	Total amount spent on the order.
payment_method	Payment method used to complete the purchase.
city	City associated with the customer’s order.
review_score	Customer satisfaction rating provided after purchase.
gender	Customer’s gender.
age	Customer’s age in years.

Data Preparation

Before analysis was conducted, data cleaning was performed in Excel to address:

- Missing values
- Standardizing data formats
- Inconsistencies in the data
- Validating data quality

Additional details are available in the accompanying Data Cleaning Report.

Calculated fields were created to support analysis, including Revenue (Quantity × Unit Price) and Average Order Value (Total Revenue ÷ Total Orders).

Analysis Methods

Microsoft Excel was used to:

- Create PivotTables
- Calculate KPI metrics
- Generate summary statistics
- Identify trends and patterns

Tableau was used to:

- Create interactive visualizations

- Develop a KPI dashboard
- Implement user interactions for the dashboard
- Create calculated fields and filters for the dashboard

Key Performance Indicators (KPIs)

KPI	Definition
Total Revenue	Sum of revenue generated from sales
Total Orders	Total number of orders placed
Total Units Sold	Total quantity of products sold
Average Order Value	Average value spent per order
Average Customer Rating	Average customer satisfaction score

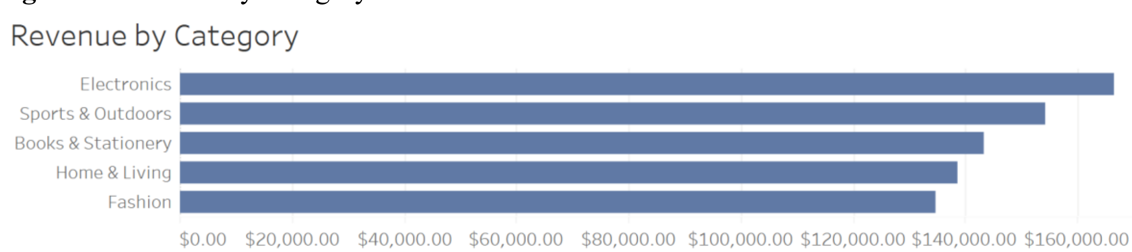
KPI Summary Results

KPI	Value
Total Revenue	$737,326.88
Total Orders	1,000
Total Units Sold	2,947
Average Order Value	$737.33
Average Customer Rating	3.19

The company generated $737,326.88 in total revenue from 1,000 orders and sold 2,947 units during the analysis period. The average order value was $737.33, indicating relatively high-value purchases. The average customer rating was 3.19 out of 5, suggesting opportunities to improve customer satisfaction and customer experience.

Results and Analysis

Figure 1. Revenue by Category



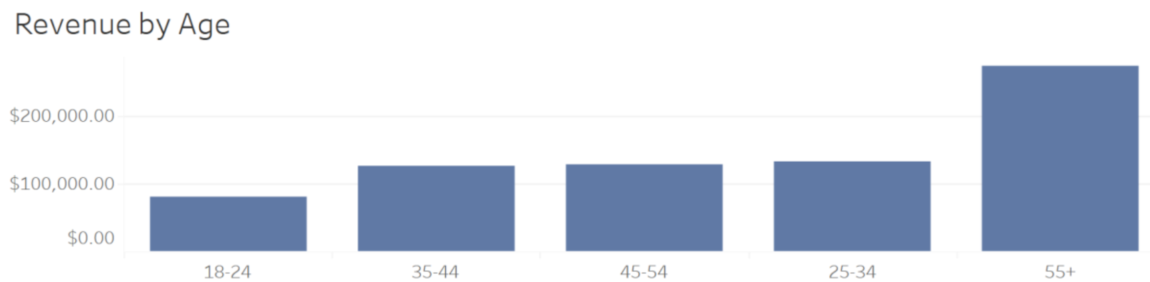
Findings:

The company sells products across five categories: Electronics, Sports & Outdoors, Books & Stationery, Home & Living, and Fashion.

Electronics generated the highest revenue at $166,510.34, making it the company's strongest-performing category. In contrast, Fashion generated the lowest revenue at $134,716.61.

This finding suggests that electronics products are a major driver of overall revenue and may represent a category where customers are willing to spend more per purchase. It also suggests that electronics is a category that should be retained. Lower-performing categories, such as Fashion, may benefit from additional marketing efforts or promotional campaigns to increase sales performance.

Figure 2. Revenue by Age Group

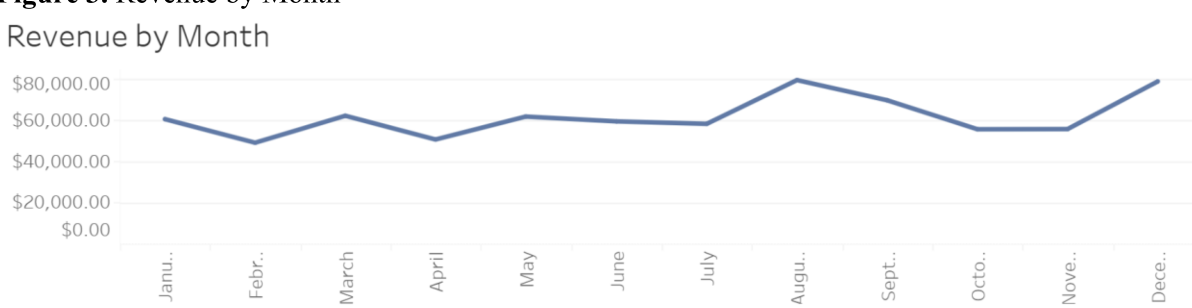


Findings:
Customers were segmented into five age groups: 18–24, 25–34, 35–44, 45–54, and 55+.

The 55+ age group generated the highest overall revenue at $271,934.93, while the 18–24 age group generated the lowest overall revenue at $81,032.66.

These results suggest that older customers may have greater purchasing power or like to purchase more frequently than younger customers. Future marketing and promotional initiatives should focus on increasing engagement among younger customers while continuing to retain high-value older customer segments.

Figure 3. Revenue by Month

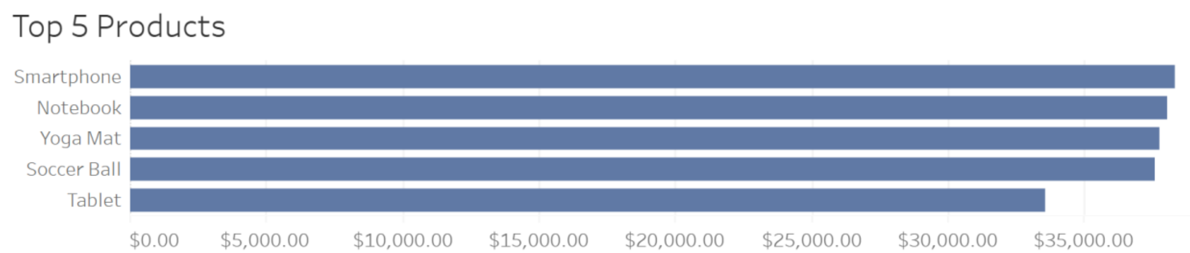


Findings:
Revenue fluctuated throughout the year of 2024 and into 2025 and exhibited recurring seasonal patterns.

Revenue generally increased during the beginning and middle months of each quarter before declining slightly towards the end of the quarter. Revenue in quarter 4 of 2024 actually saw an increase during December, when revenue reached approximately $78,000, exceeding revenue levels observed during other quarter-end months.

This trend suggests that seasonal purchasing behavior may influence sales performance. Creating promotional campaigns during seasons and holidays could possibly maximize revenue during peak periods and maybe improve performance during slower periods.

Figure 4. Top Products

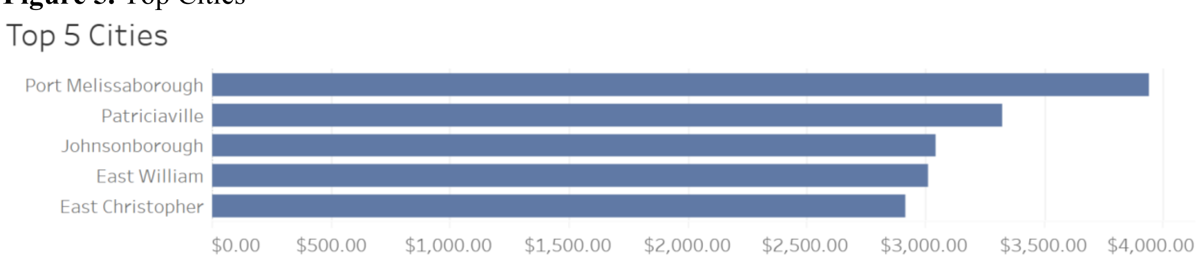


Findings:
The top 5 performing products by revenue between 2024 and 2025 were:

- Smartphone at $38,319.26
- Notebook at $38,027.68
- Yoga Mat at $37,752.08
- Soccer Ball at $37,587.30
- Tablet at $33,581.02

These products generated the highest revenue across the product offerings and represent important contributors to overall revenue performance. Further analysis on customer purchasing behavior and characteristics with high-performing products may help guide future inventory planning and marketing decisions.

Figure 5. Top Cities



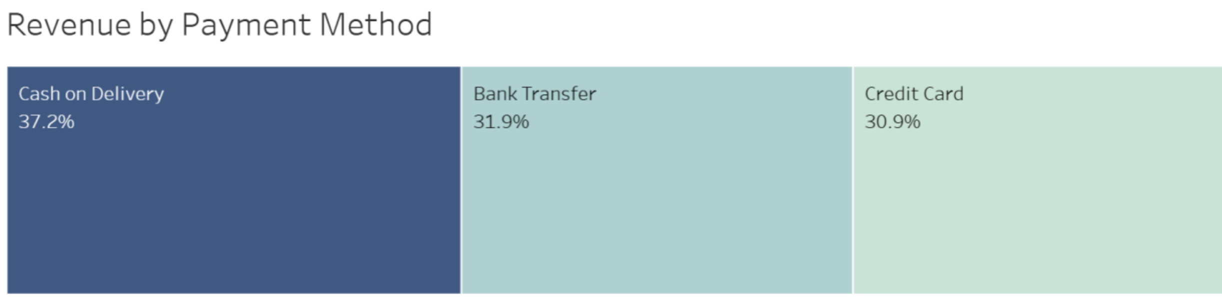
Findings:
The cities in this dataset are fictional and are only intended for demonstration purposes.

Out of all the cities represented in the dataset, the top 5 revenue-generating cities were:

1. Port Melissaborough at $3,941.29
2. Patriciaville at $3,324.10
3. Jacksonborough at $3,045.09
4. East William at $3,011.36
5. East Christopher at $2,919.20

Although the locations are fictional, the analysis demonstrates how geographic sales analysis can be used to identify high-performing markets and support regional marketing strategies.

Figure 6. Revenue by Payment Method



Findings:
Customers used the following payment methods:

- Cash on Delivery, accounting for 37.2% of total sales.
- Bank Transfer, accounting for 31.9% of total sales.
- Credit Card, accounting for 30.9% of total sales.

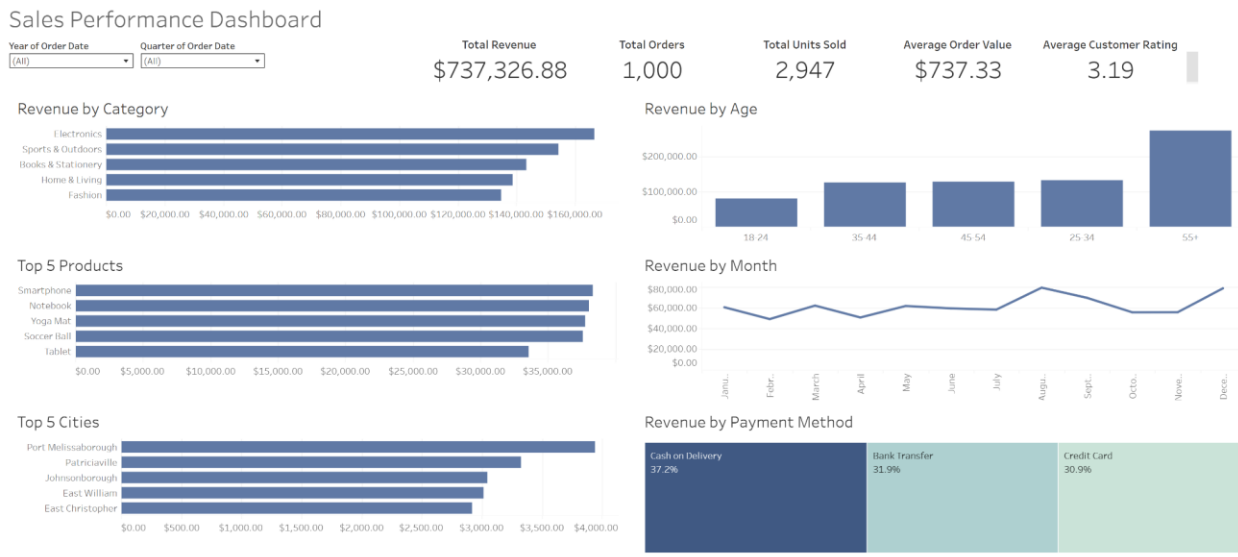
Cash on Delivery was the most frequently used payment method, accounting for over one-third of all sales. This suggests that customers like having multiple payment options when completing purchases.

Dashboard Overview
An interactive Tableau dashboard was developed to allow users to explore sales performance by year and quarter.

Dashboard Features

- KPI summary card
- Revenue trend analysis
- Product performance analysis
- Customer demographic analysis
- Geographic analysis
- Payment method analysis

- Interactive filters



<u>Key Findings</u>

The analysis produced the following key findings:

1. Electronics is the strongest-performing product category.
2. Fashion and Home & Living generated lower revenue than other categories.
3. Customers aged 55+ contributed the largest share of revenue.
4. Customers aged 18–24 contributed the smallest share of revenue.
5. Revenue exhibited seasonal fluctuations throughout the year.
6. Cash on Delivery was the most commonly used payment method.

<u>Business Recommendations</u>

Recommendation 1: Expand Seasonal Promotions

Revenue trends indicated seasonal fluctuations throughout the year, with stronger performance during the beginning and middle months of the quarter. Implementing targeted promotions during slower periods may help increase revenue and increase customer purchasing activity. Promotional efforts should focus particularly on lower-performing categories such as Fashion and Home & Living, but should still retain the high-performing categories.

Recommendation 2: Analyze Product Ratings and Customer Satisfaction

Although customer review scores were included in the dataset, the analysis was limited to only revenue performance. Future analysis should examine relationships between customer ratings and product sales performance to identify the products that generate strong revenue but receive lower customer satisfaction scores. These insights could support product improvement initiatives and customer retention efforts.

Recommendation 3: Increase Marketing Efforts Toward Younger Customers

Customers aged 18–24 generated the lowest revenue among all age groups. Targeted marketing campaigns, promotional offers, and customer engagement initiatives focused on younger consumers may help increase sales within this segment while expanding the customer base.

Project Limitations

The following limitations should be considered when interpreting results:

- The dataset represents a fictional company.
- Customer locations are fictional.
- The analysis only covers data from the end of Q1 2024 to Q1 2025.
- Website data, such as items still in customers' carts, what items customers click on, and discounts applied, could not be analyzed.
- Written customer feedback was not available for review.

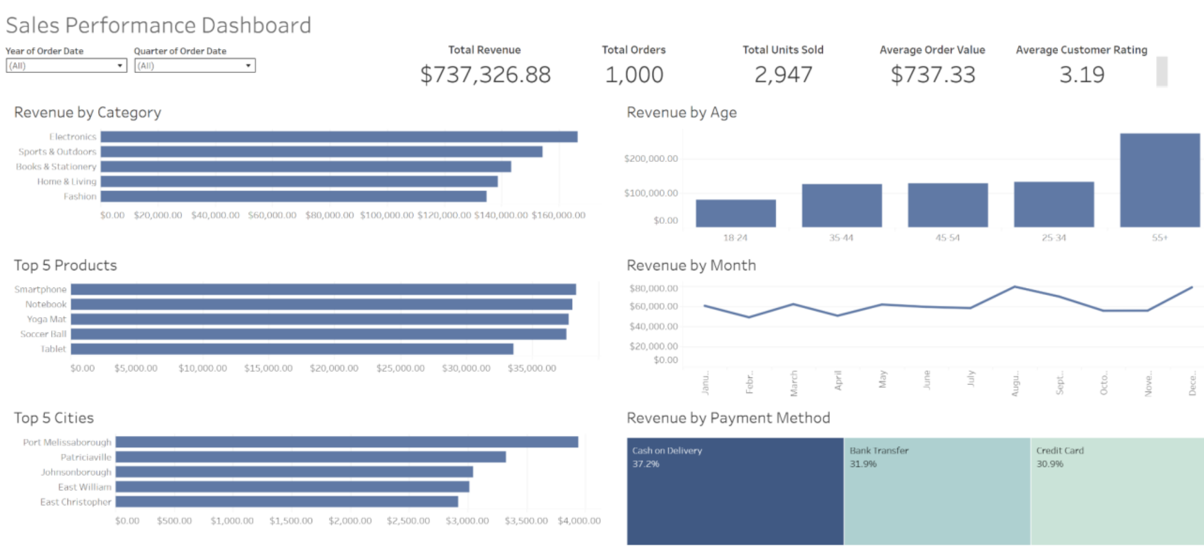
Conclusion

This analysis identified how revenue performs across products, customer ages, cities, and payment methods. Electronics showed to be the highest-performing category, customers aged 55+ generated the highest revenue, and revenue displayed seasonal trends throughout 2024 and 2025.

The Tableau dashboard provides an interactive view of revenue performance and allows users to explore sales metrics across multiple dimensions. The findings and recommendations offer insights into improving marketing strategies, customer targeting, and inventory planning.

Appendix

Appendix A – Dashboard Screenshot



Appendix B – Supporting PivotTables

Figure B1. Revenue by Category PivotTable

Years (order_date)	(All)		
Quarters (order_date)	(All)		
Revenue by Category			
Row Labels	**Sum of revenue**	**Average of price**	**Sum of quantity**
Electronics	$166,510.34	$259.05	648
Sports & Outdoors	$154,346.26	$251.02	625
Books & Stationery	$143,215.52	$261.07	547
Home & Living	$138,540.15	$243.18	563
Fashion	$134,714.61	$244.59	564
Grand Total	**$737,326.88**	**$251.85**	**2947**

Figure B2. Revenue by Product PivotTable

Years (order_date)	(All)
Quarters (order_date)	(All)
Top 5 Products	
Row Labels	**Sum of revenue**
Smartphone	$38,319.26
Notebook	$38,027.68
Yoga Mat	$37,752.08
Soccer Ball	$37,587.30
Tablet	$33,581.02
Grand Total	**$185,267.34**

Years (order_date)	(All)
Quarters (order_date)	(All)
Bottom 5 Products	
Row Labels	**Sum of revenue**
Carpet	$24,175.74
Tent	$23,449.52
Pen	$23,273.07
Novel	$22,750.81
Shirt	$18,230.94
Grand Total	**$111,880.08**

Figure B3. Revenue by City PivotTable

Years (order_date)	(All)	
Quarters (order_date)	(All)	
Top 5 Cities		
Row Labels	**Sum of revenue**	**Sum of orders**
Port Melissaborough	$3,941.29	86855
Patriciaville	$3,324.10	189049
Johnsonborough	$3,045.09	107008
East William	$3,011.36	163644
East Christopher	$2,919.20	129810
Grand Total	**$16,241.04**	**676366**

Years (order_date)	(All)	
Quarters (order_date)	(All)	
Bottom 5 Cities		
Row Labels	**Sum of revenue**	**Sum of orders**
Gonzalezshire	$27.65	34358
Port Kimberlymouth	$24.00	32747
Williamton	$22.21	87478
North Diane	$21.44	99243
Jonathanstad	$20.84	74178
Grand Total	**$116.14**	**328004**

Figure B4. Revenue by Age Group PivotTable

Years (order_date)	(All)
Quarters (order_date)	(All)
Revenue by Age Group	
Row Labels	**Sum of revenue**
18-27	$126,748.87
28-37	$116,747.34
38-47	$124,850.56
48-57	$141,288.05
58-67	$134,322.96
68-77	$93,369.10
Grand Total	**$737,326.88**

Figure B5. Revenue by Month PivotTable

Revenue by Month	
Row Labels	Sum of revenue
2024	**597071.69**
Qtr1	**30620.98**
Mar	30620.98
Qtr2	**170950.98**
Apr	50375.4
May	61448.18
Jun	59127.4
Qtr3	**206333.07**
Jul	57939.86
Aug	79070.12
Sep	69323.09
Qtr4	**189166.66**
Oct	55328.98
Nov	55404.74
Dec	78432.94
2025	**140255.19**
Qtr1	**140255.19**
Jan	60210.83
Feb	48826.04
Mar	31218.32
Grand Total	**737326.88**

Figure B6. Revenue by Payment Method PivotTable PivotTable

Years (order_date)	(All)
Quarters (order_date)	(All)

Revenue by Payment Method	
Row Labels	**Sum of revenue**
Cash on Delivery	$274,375.64
Bank Transfer	$234,853.69
Credit Card	$228,097.55
Grand Total	**$737,326.88**

Appendix C – Data Cleaning Report
See the accompanying Data Cleaning Report document in the retail-sales-data repository for detailed data preparation procedures.

Appendix D - Original Kaggle Dataset
https://www.kaggle.com/datasets/ertugrulesol/online-retail-data

Appendix E - GitHub Repository
https://github.com/naguthery/Ecommerce-Sales-Analytics